Inverse variation

1 $y = \frac{k}{x}$

2 $k = 150$

3

a Direct

b Inverse

4 $r = \frac{k}{a}$

5

a No

b No

c No

d Yes

e Yes

f No

6

a 375

b $\frac{125}{2}$

c $\frac{125}{4}$

7

a Yes

b Yes

c No

d No

8

a $k = 252$

b $x = 9, y = 42$

9

a $t = \frac{48}{s}$

b $t = \frac{2}{s^2}$

10

a 0

b 0

Key features of hyperbola

11

a 0

b ∞

c $x = 0$

d $y = 0$

e $y = x, y = -x$

12

a -4

b Negative

c Positive

d 2 and 4

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13

a

x	-2	-1	$-\frac{1}{2}$	$-\frac{1}{10}$	$-\frac{1}{100}$	$\frac{1}{100}$	$\frac{1}{10}$	$\frac{1}{2}$	1
y	-1	-2	-4	-20	-200	200	20	4	2

b 0

c $x = \frac{2}{y}$

d 0

e As x approaches 0 from the right, the function value approaches ∞. As x approaches 0 from the left, the function value approaches $-\infty$.
As x approaches ∞ and $-\infty$, y approaches 0. This is called the limiting value of the function.

14

a 0 **b** $-\infty$ **c** ∞ **d** 0

e 0

15 6

16

a Yes **b** No **c** No

17

a $x = 0$ **b** $y = 0$

c All real values of x, except $x = 0$. **d** All real values of y, except $y = 0$.

18 The function will be undefined.

19

a $x = 0$ only. **b** All real values of x, except $x = 0$.

c $y = 0$ **d** All real values of y, except $y = 0$.

20 Looking at $y = \frac{2}{x}$, notice that the numerator 2 is non-zero, and so $\frac{2}{x}$ can never be equal to 0.
Another approach is to rearrange the equation and make x the subject. Then we get $x = \frac{2}{y}$.
In this form we can see that the denominator y cannot be equal to 0.

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21

a 2

b 1

c $y = \frac{4}{x}$

d The graph moves further from the axes

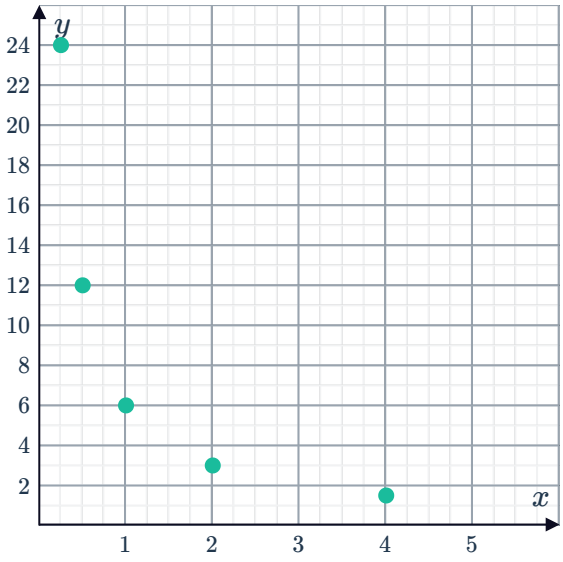
Graphs of hyperbolas

22

a

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y	24	12	6	3	$\frac{3}{2}$

b



23 She can label a point on the graph.

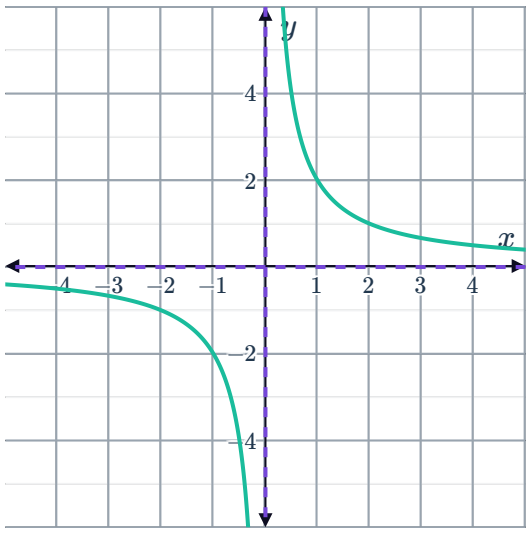
24

a

i

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	-1	-2	-4	4	2	1

ii



iii 1 and 3

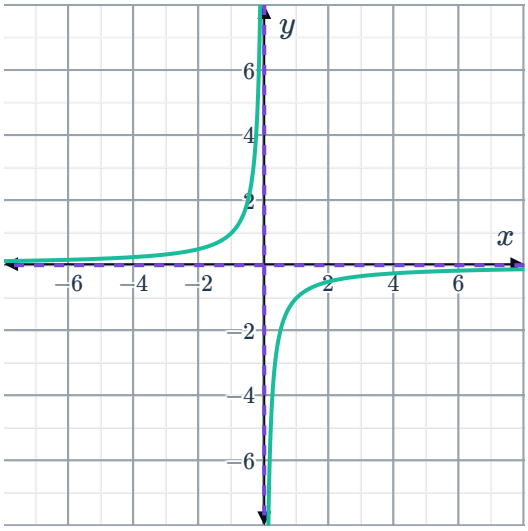
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b

i

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	$\frac{1}{2}$	1	2	-2	-1	$-\frac{1}{2}$

ii



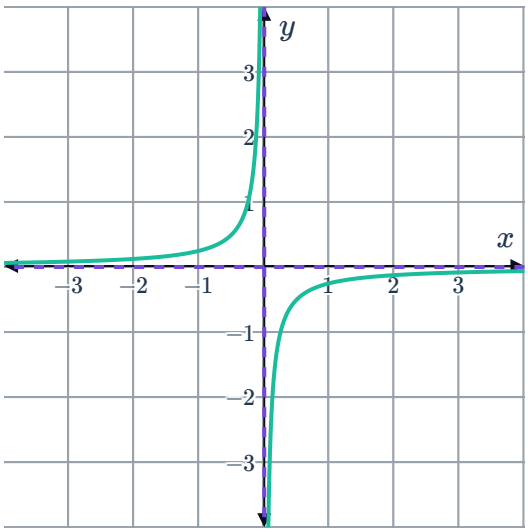
iii 2 and 4

c

i

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{4}$	$-\frac{1}{8}$

ii



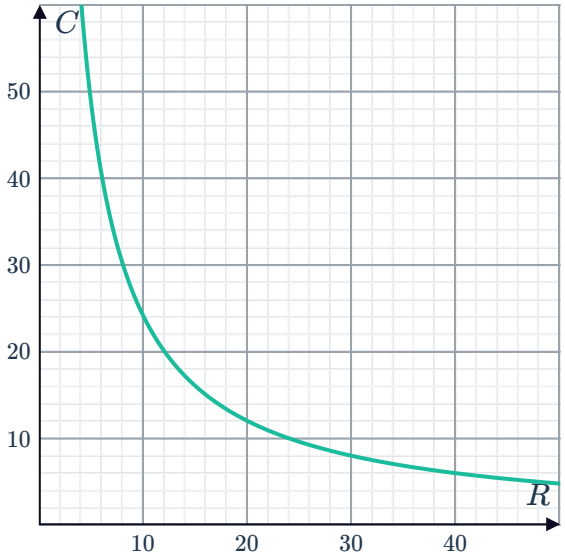
iii 2 and 4

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Applications

25

a



b The current decreases.

26

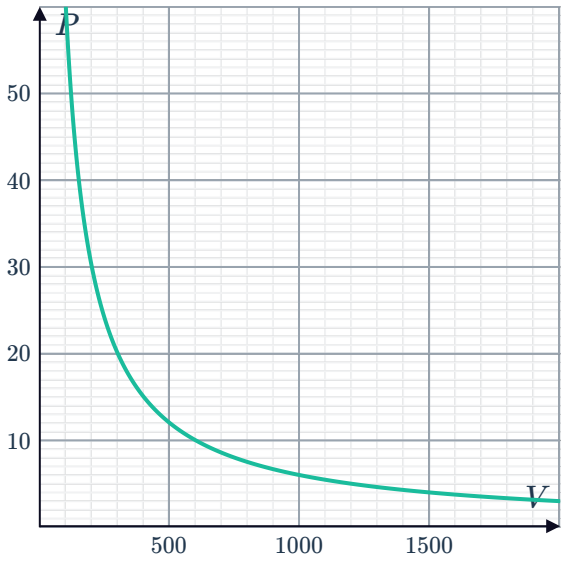
a -2

b 32 units^2

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27

a

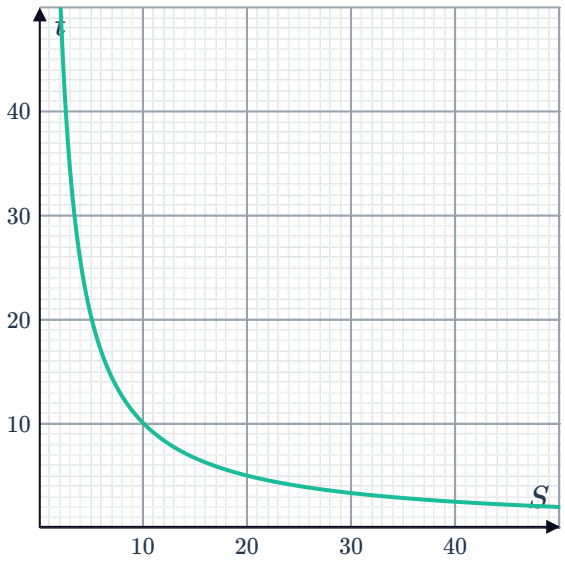


b $6000\ \text{kg/cm}^2$

c The pressure decreases.

28

a



b 10 hours

c 2 hours

d The speed must increase.

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